

HealthConnect Clinic

Week 8 — Final Analytics & Decision Support Package
Data Analytics Track | AnalystLab Africa Experience Lab

Project question: How can HealthConnect Clinic use data and AI to reduce missed appointments and improve the patient support experience?

Scope of this package: the final, decision-ready output of the Data Analytics track — validated KPIs, the final risk-triage dashboard, business insights, recommendations, and the closed cross-track validation loop with Data Science. This package builds on, and does not repeat, the Week 5–7 analysis and testing.

EXECUTIVE SUMMARY

HealthConnect's own appointment data confirms that a small number of factors reliably predict missed appointments: how far in advance a patient books, how far they live from the clinic, and whether they've missed appointments before. Combining these into a simple 0–3 risk score identifies just over half of all appointments (54.4%) that account for nearly two-thirds of all no-shows (62.9%) — a far more effective triage tool than any single factor alone. Testing confirms this score is accurate, reproducible, and works consistently across patient groups. SMS reminders help all patients by a similar amount and should stay universal rather than targeted by risk. An independent model built by the Data Science track agrees with this analysis and offers a modest further improvement once integrated into HealthConnect's systems. Together, these findings give HealthConnect a validated, ready-to-use way to focus limited staff time on the patients most likely to need it.

Week 7 → Week 8 Transition

Item	Detail
Main Week 7 output	The Week 7 Data Analytics Testing & Refinement Report and notebook — reproducibility testing, segment-stability testing, a properly powered reminder test, and cross-track validation against Data Science's model
Most important testing result	Confirmed, with adequate statistical power, that the reminder effect does not vary by risk tier, and validated the compound risk score's stability across patient segments (12 of 13)
Major issue discovered	A headline KPI (59.0% no-show rate for the flagged segment) was a calculation error — an unweighted average instead of the correct n-weighted pooled rate; corrected to 56.0%
What was refined/improved	The headline KPI, the reminder test's statistical power (split-sample → pooled interaction model), and the cross-track benchmark (interim proxy → Data Science's actual model)
Component now ready	The compound risk score and its supporting KPIs/dashboard are tested, corrected, and validated — ready for final packaging and an operational recommendation
What still needs to be integrated	Coordination with ML Engineering on how the risk score and the Data Science model will coexist in the operational workflow; the reminder-timing/channel A/B test design
Track(s) involved in final integration	Data Science (model benchmark), Project Management (final coordination), ML Engineering (indirectly, for future pipeline design)
What I intend to demonstrate	How testing turned an initially flawed headline figure and an under-powered test into a validated, cross-checked analytics deliverable that directly informs HealthConnect's patient outreach strategy

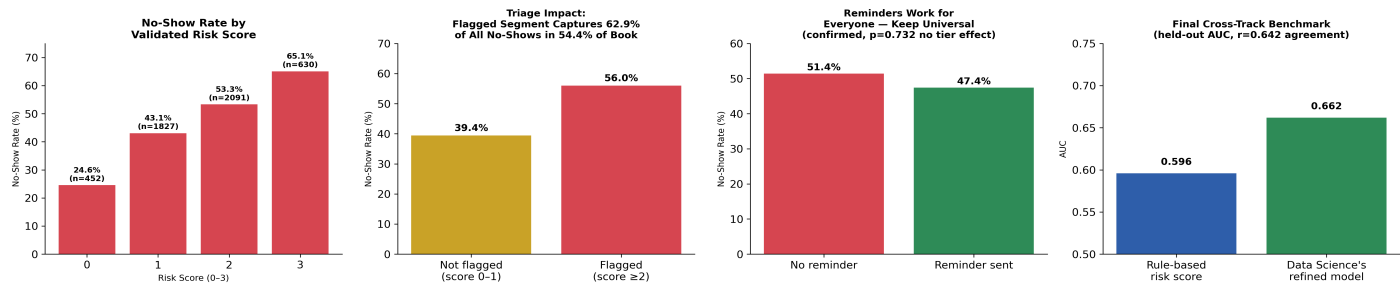
Part 1 — Final Integration Readiness

Requirement	Response
What is my final component?	The compound no-show risk score (0–3), the validated statistical driver findings, and the final analytics dashboard/report
What problem does it address?	Identifying which patients are likely to miss appointments, so staff can prioritise limited outreach capacity
What is its current status?	Tested, refined, validated across patient segments, and cross-checked against Data Science's model — ready for operational use
What did I improve during Week 7?	Corrected a headline KPI (59.0%→56.0%), upgraded the reminder finding to adequately powered/confirmed, validated segment stability, and closed the cross-track validation loop
Which other track(s) does it depend on?	Data Science (model benchmark); Project Management (coordinating the reminder-timing test and rollout)
Which track(s) depend on my output?	Data Science (feature validation, benchmark); Project Management (business recommendations)
What output will I provide?	Final risk-score definition and thresholds, the validated driver list, the final dashboard, KPI figures, and business recommendations
What output do I need from another track?	Data Science's finalised model documentation (received); ML Engineering's plan for how the model will be operationalised (for Week 8 coordination)

Requirement	Response
What evidence demonstrates readiness?	The Week 7 testing & validation record, corrected and reproducible KPIs, 12/13 segment stability, adequately powered statistical tests, and cross-track AUC validation
What limitations remain?	Synthetic dataset, heuristic thresholds, small subgroup sample sizes, and the reminder-timing test not yet run

Final Analytics Dashboard

HealthConnect Clinic — Week 8 Final Analytics Dashboard



Panel 1: the validated, reproducible risk-score gradient. Panel 2: the operational impact of the risk score as a triage filter. Panel 3: reminders work broadly the same for everyone and should remain universal. Panel 4: the final, held-out cross-track benchmark between the rule-based score and Data Science's model.

Final KPIs

KPI	Value
Overall no-show rate (all appointments)	48.5%
No-show rate — patients not flagged (risk score 0–1)	39.4%
No-show rate — patients flagged (risk score ≥2)	56.0%
Share of appointments flagged by the risk score	54.4%
Share of all no-shows captured by the flagged segment	62.9%
No-show rate — reminder sent	47.4%
No-show rate — no reminder sent	51.4%
Cross-track benchmark — rule-based score (held-out AUC)	0.596
Cross-track benchmark — Data Science's model (held-out AUC)	0.662

Validated Findings & Business Insights

- **Three independent, statistically validated drivers:** booking lead time, distance to clinic, and prior no-show history each remain significant predictors even after controlling for the other two — confirmed via chi-square testing, joint logistic regression, and (independently) Data Science's own feature-importance analysis.
- **Waiting time and appointment day are not drivers** — confirmed consistently across every testing round in both the Data Analytics and Data Science tracks.
- **The compound risk score (0–3) is validated and stable** — reproducible after the Week 7 KPI correction, and its monotonic gradient holds across patient gender, age group, and appointment type (12 of 13 segments tested).
- **Reminders are effective and should remain universal** — a properly powered test (Week 7) confirms the reminder effect does not vary meaningfully by risk tier, so there is no analytical case for restricting reminders by risk.
- **Data Science's model adds real, if modest, value on top of the rule** — a held-out AUC of 0.662 vs. 0.596 for the simple rule, and the two approaches agree directionally ($r = 0.642$). The model is best used as a complement to, not an immediate replacement for, the interpretable rule-based score.

Recommendations

- Deploy the compound risk score (0–3) as the primary administrative triage tool: light-touch (reminder only) for scores 0–1, proactive outreach for score 2, and priority phone-call/rescheduling support for score 3.
- Keep SMS reminders universal across all patients, regardless of risk tier.
- As ML Engineering integrates Data Science's model into a production pipeline, treat it as a secondary refinement layer on top of the rule-based score rather than an immediate replacement — the rule is simpler to explain to clinic staff and easier to audit, while the model adds incremental predictive lift once operationalised.
- Track these KPIs going forward: overall no-show rate; no-show rate and capture by risk tier; SMS reminder compliance; and model-vs-rule agreement (to monitor drift over time).
- Prioritise the stratified reminder-timing/channel test as the next analytical investment, now that tier-based targeting of reminders has been ruled out as a design factor.

Analytical Limitations

- The dataset is fictional/synthetic; all findings require validation against real HealthConnect operational data before deployment.
- Findings are observational (association, not causation).
- Risk-score thresholds remain heuristic, chosen by testing a small grid of options rather than formally optimised; Data Science's model is the more rigorous, data-driven alternative for future refinement.
- Some patient subgroups (e.g. the smallest gender category, certain age bands at the highest risk tier) are too small for fully reliable segment-level conclusions.

- No live A/B test has yet been run on reminder timing or channel — this remains a recommended next step, not a completed analysis.

HC-POD Final Integration

Item	Detail
Track collaborated with	Data Science (primary); Project Management (secondary, for final coordination)
Dependency	Data Science needs Data Analytics' validated features and risk score as a benchmark; Project Management needs final business recommendations for project closure
Output received	Data Science's final refined model (Logistic Regression, 5 features), the held-out AUC benchmark (0.596 rule-based / 0.662 model-based), and a methodological note on in-sample vs. held-out evaluation
Output provided	The corrected headline KPI (56.0%), the confirmed reminder finding, and the validated risk score and driver list
Final integration activity	The Week 7 cross-track benchmark comparison (Test E) was finalised and carried into this Week 8 dashboard/report as the reference figures for both tracks
What changed	Both tracks now cite one shared, held-out-validated benchmark instead of two separate (and initially inflated, in-sample) figures
How the change improved the solution	Gives HealthConnect a single, defensible set of numbers for both the interpretable rule and the model, supporting a two-layer triage design: the rule for transparency and auditability, the model for incremental predictive lift
Evidence	Week 7 Test E (both tracks' notebooks/reports); this Week 8 final dashboard
Contribution to final presentation	The cross-track validation section anchors the "how the components work together" part of the HC-POD walkthrough and this track's individual video presentation

Executive Summary (Standalone)

HealthConnect can identify roughly half of its appointment book as meaningfully higher-risk using three factors already in its own data — booking lead time, distance to the clinic, and prior no-show history. This risk score, now tested and validated, captures nearly two-thirds of all missed appointments and should anchor a tiered outreach strategy. SMS reminders help broadly and should stay universal. A complementary predictive model from the Data Science track offers a modest additional lift and is ready for phased integration once ML Engineering completes the operational pipeline.